AtPoE

Artificial Intelligence Admitting the Possibilities of Error

Kirsten Lavers and Peter Murray-Rust

Kirsten Lavers’ drawings speak to many areas of science and mathematics. This very informal workshop will explore these ideas and help Kirsten and Peter refine their ideas and popularise them outside the artistic world.

The audience is an integral part.

No knowledge of science is required
Wikipedia links provided

Programme

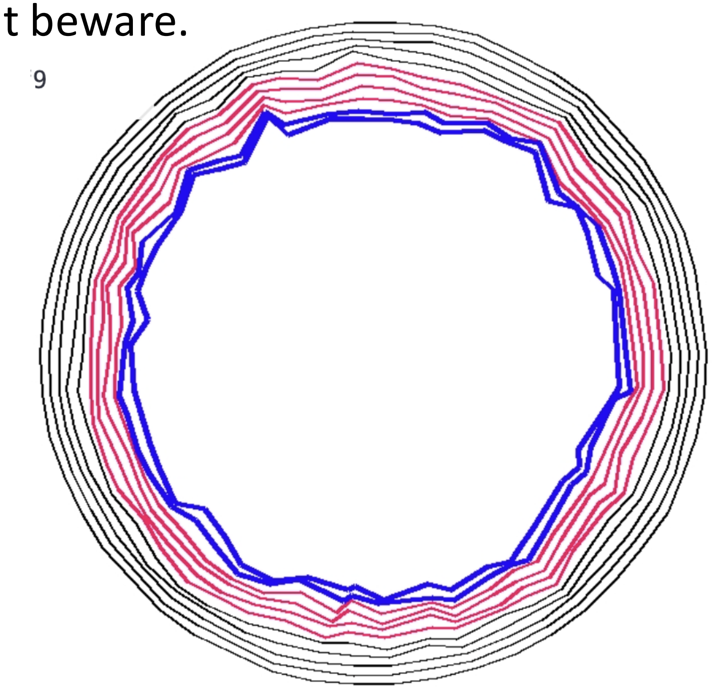
- Introductions including audience
- PMR vision of KL drawings
- Algorithms (short practical session)
- What is AI and vibe-coding? We write an algorithm
- KL conversation with ChatGPT
- Display of AtPoE pattern generator program
- Discussion

WARNING ! : AIs are NOT sentient*

Human phrases are programmed into them.

- “You’re absolutely right!”
- “Let me implement the plotting”

I shall converse with them as if they are human, but beware.



The use of AIs is contentious.

We might discuss:

- extractive surveillance capitalism
- Brain rot of frequent users
- Carbon budgets

* https://en.wikipedia.org/wiki/Stochastic_parrot

https://www.c2d3.cam.ac.uk/events/ai-con-conversation-professor-emily-m-bender

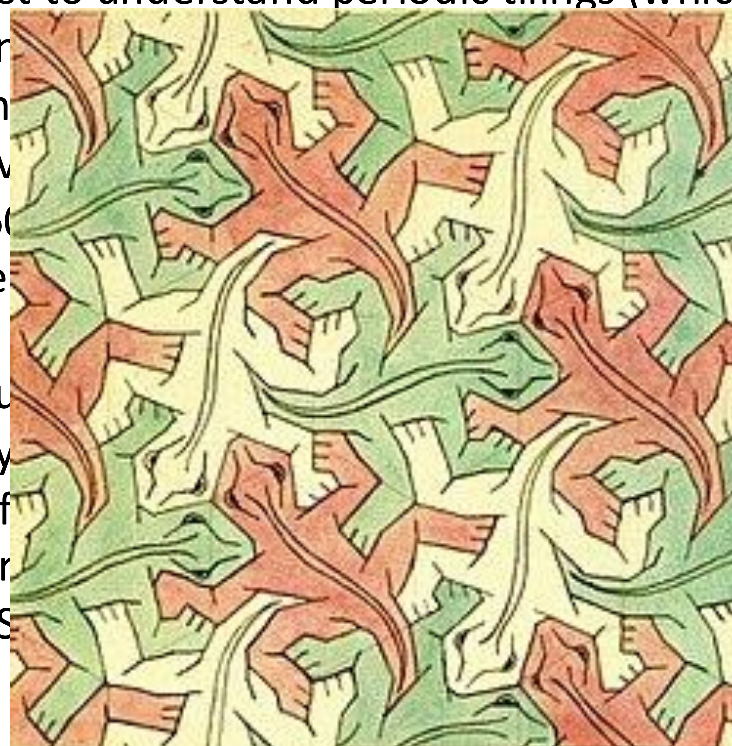
Scientific Topics related to AtPoE

- Emergence
- Creation
- Crystallisation and domains
- Instability
- Random Error
- Error propagation
- Symmetry breaking
- Iteration
- Algorithms
- Chaos
- Morphogenesis
- Cellular Automata
- Chirality
- Layers
- Molecular dynamics

Crystallography and Escher

Forty-five years ago [1960] , a relatively unknown Dutch graphic artist, M.C. Escher, gave a standing-room-only lecture to the Fifth International Congress of the IUCr in Cambridge, England. There was an accompanying exhibit of his work that amazed the crystallographers. His pioneering work in exploring colour symmetry was a rare instance of an artist investigating a field before "official crystallography even thought about [it]."[1] Escher's quest to understand periodic tilings (which he called

'regular divisions of the plar … icles in
Zeitschrift für Kristallograph … phers
(notably, Caroline MacGillav … ought him out to
learn from his work. In 196 … M.C. Escher
contained a crystallographe … metry groups.
In 1965, the IUCr published … rated with Escher.
We discuss how Escher's qu … ured periodic tilings
differed from that of the cry … some of his original
investigations are worthy of … avry C.H., Symmetry
Aspects of M.C. Escher's Per … 2] Schattschneider
D., M.C. Escher: Visions of S … 1990, Harry Abrams,
New York, 2004. Keywords: … lographic teaching



International Union of Crystallography 2005

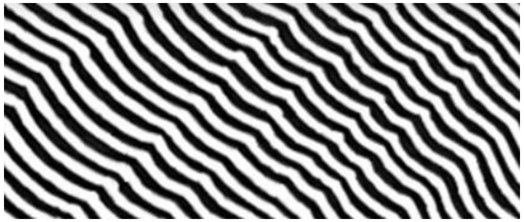
Kirsten's Algorithm (simple version)

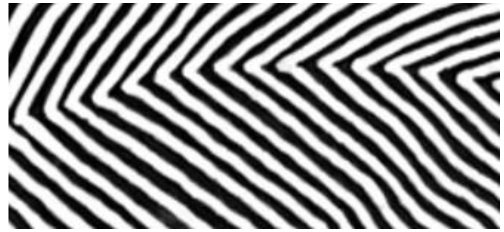
- Step0: start with a perfect circle (the boundary polygon) containing a void (normally white)
- Step1: start at a position about a pencil-lead width inside the boundary
- Rule2: draw with the pencil keeping "close to the boundary" * creating new boundary
- Repeat Step1 and Step2 until you are bored or have filled the space

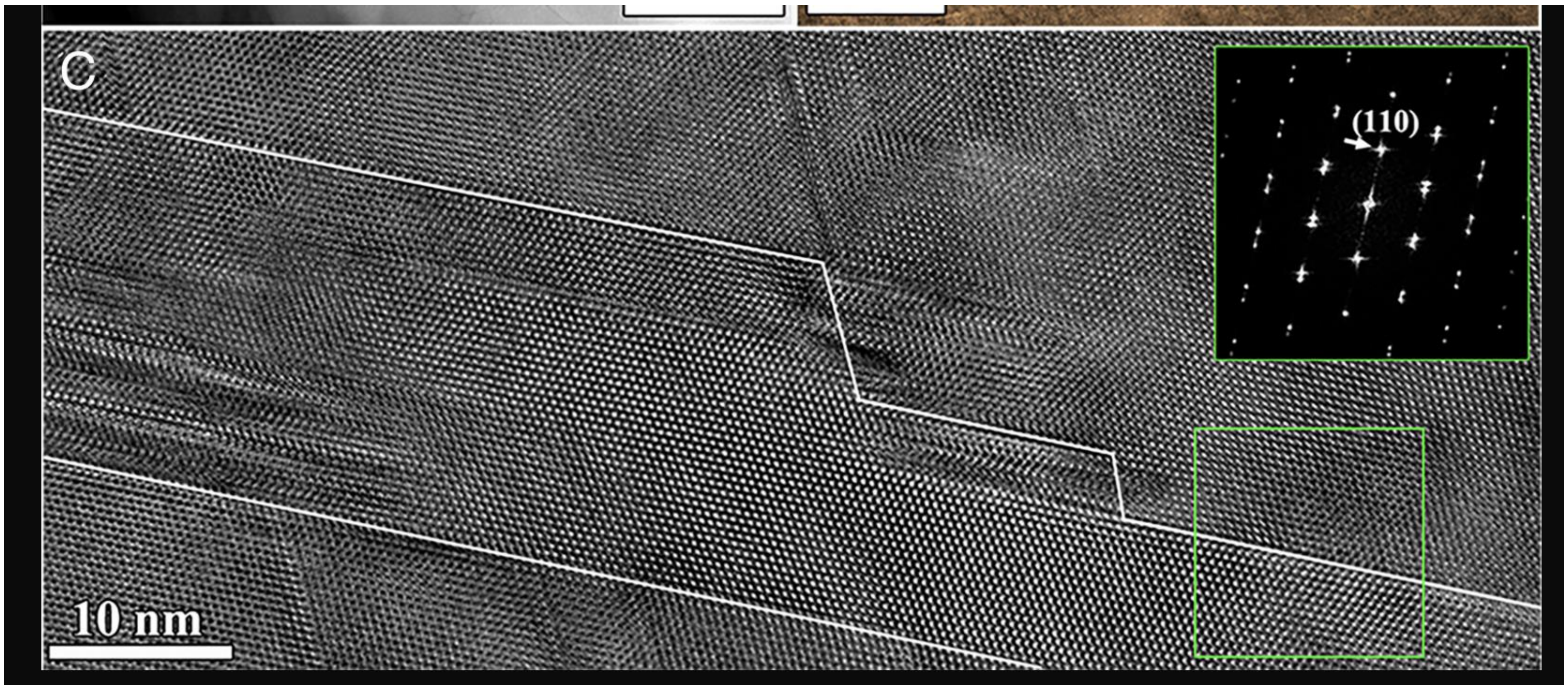
* Not too close, not too far. Just right

Grains and Domains

https://commons.wikimedia.org/wiki/File:Ammonium_Chloride_Crystals.jpg



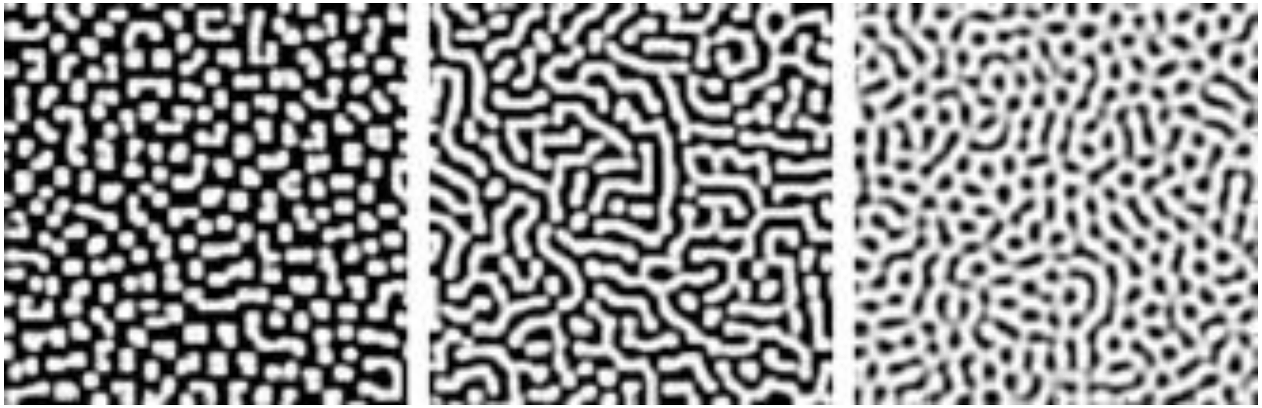


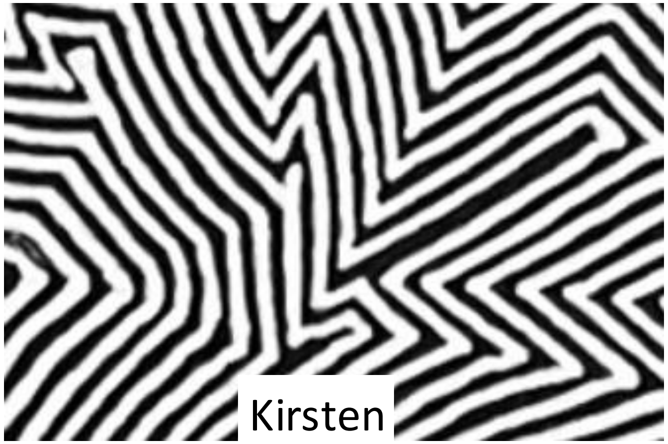


Uncovering the crystal defects within aragonite $CaCO_3$

Xingyuan San https://orcid.org/0000-0001-7975-7185, Mingyu Gong https://orcid.org/0000-0002-8120-5264, Jian Wang https://orcid.org/0000-0001-5130-300X jianwang@unl.edu, **+4** , and Xiaobing Hu https://orcid.org/0000-0002-9233-8118 xbhu@northwestern.eduAuthors Info & Affiliations

Morphogenesis

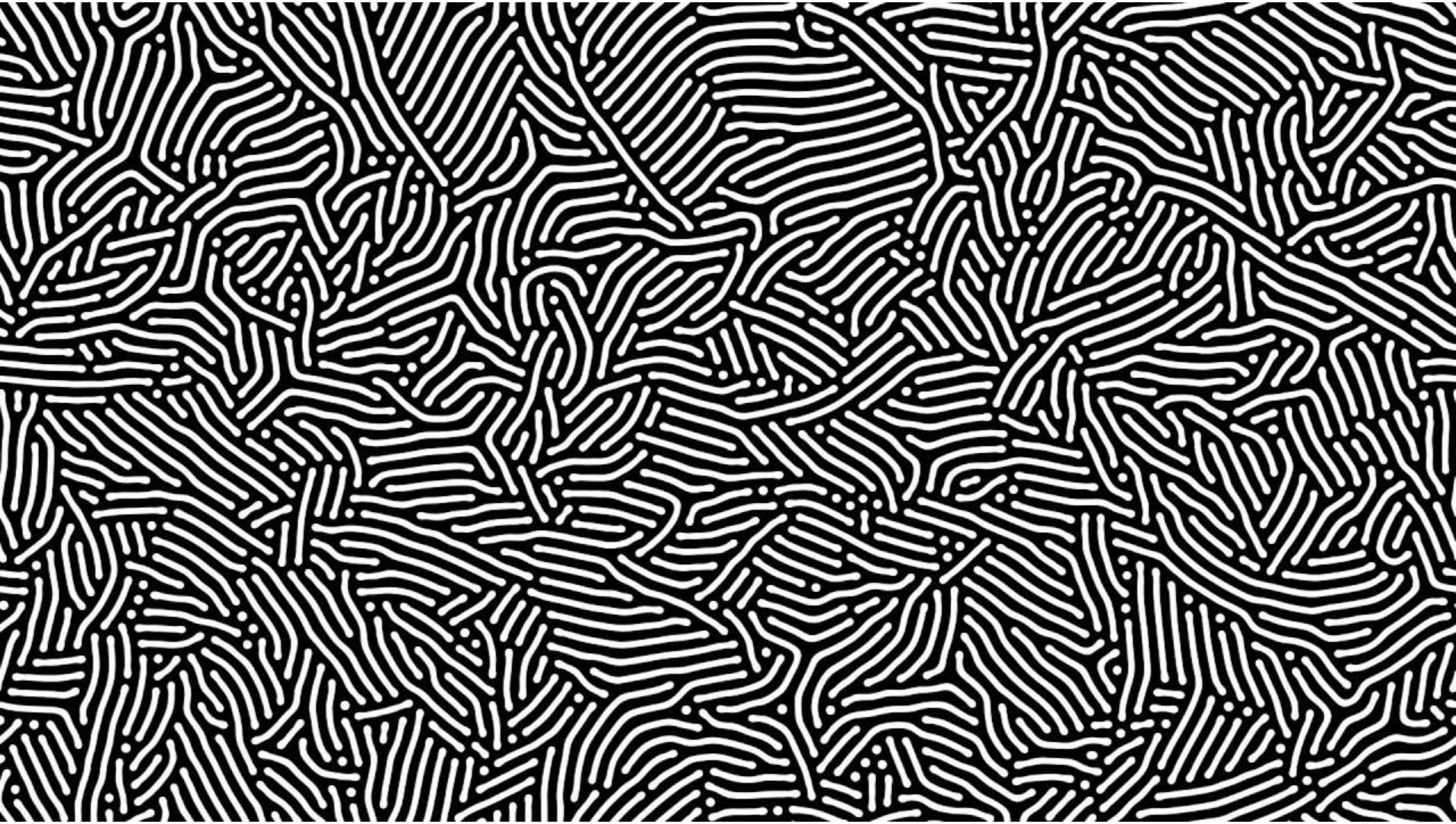




https://en.wikipedia.org/wiki/Turing_pattern

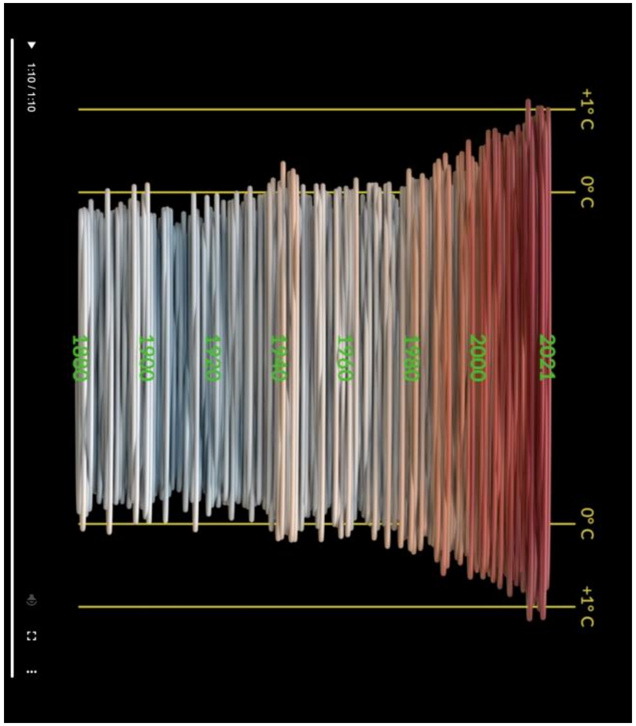
https://en.wikipedia.org/wiki/Reaction%E2%80%93diffusion_system#Two-component_reaction%E2%80%93diffusion_equations

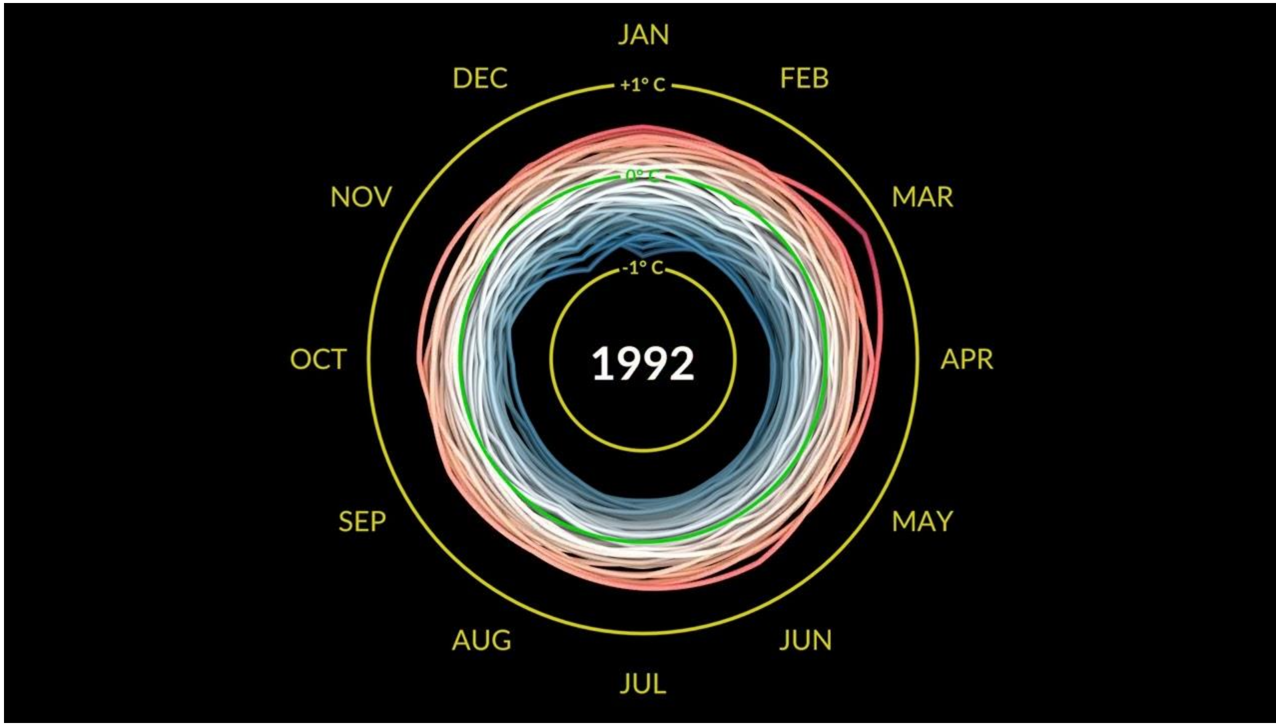
Diffusion-reaction system by P. Gray and S. K. Scott



https://youtu.be/GB7-OQk-fFk?si=kOZvXDeCuHCc5myc

Climate Spiral

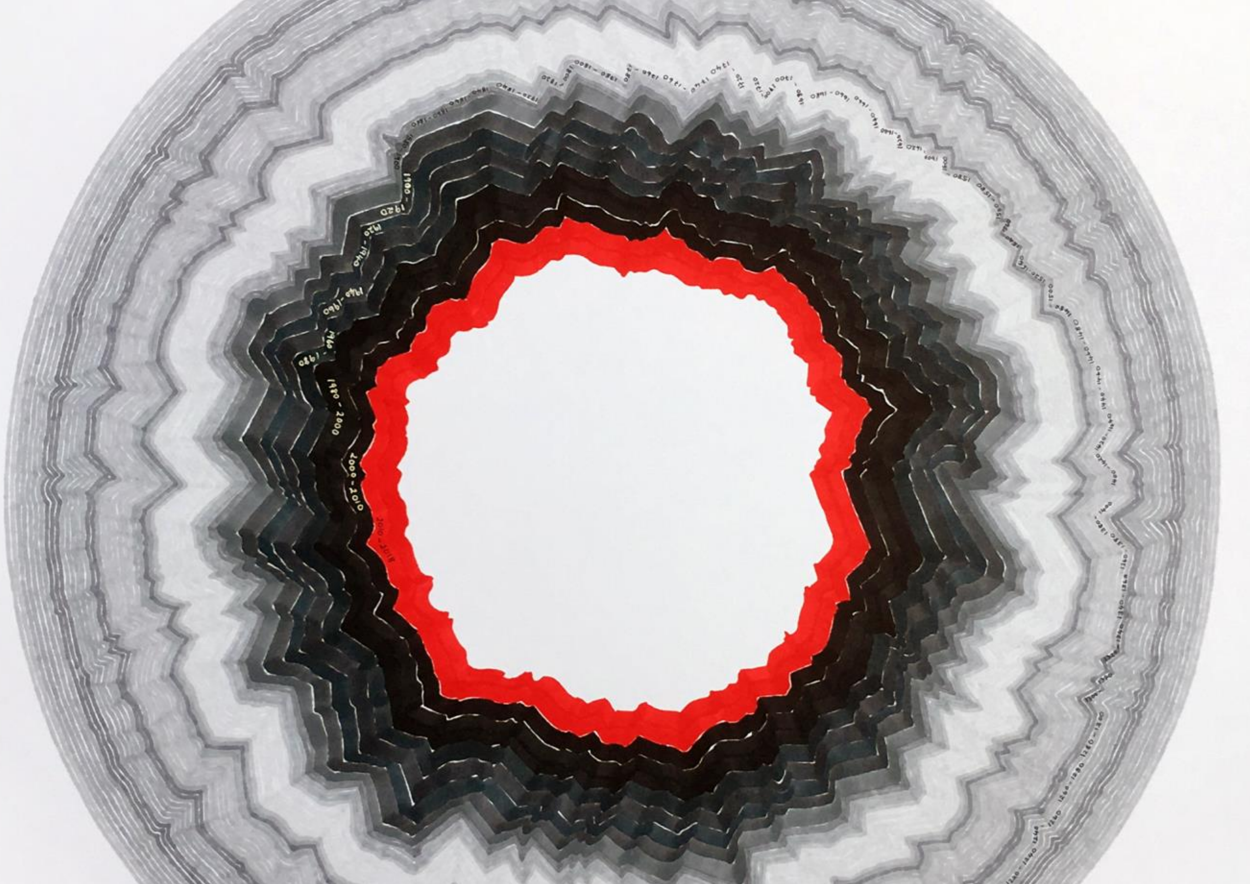




Temperature vs Date

Climate Spiral animation

https://youtu.be/jWoCXLuTIkI?si=beOwcBUKVbe6eLaG



ADMITTING THE POSSIBILITIES OF ERROR – GLOBAL CO2 LEVELS 1000 – 2018 A.D.

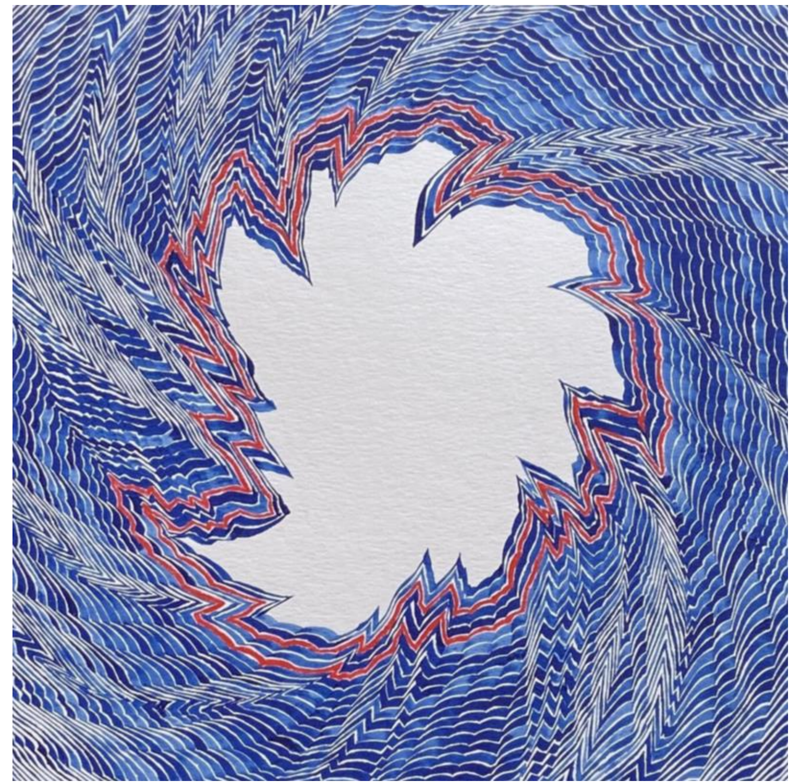
275–279	279–283	283–287	287–291	291–295	295–299	299–303	303–310	310–319	319–355	355–365	365–415

CARBON DIOXIDE CO2 PPM

DATA FROM 2DEGREES INSTITUTE

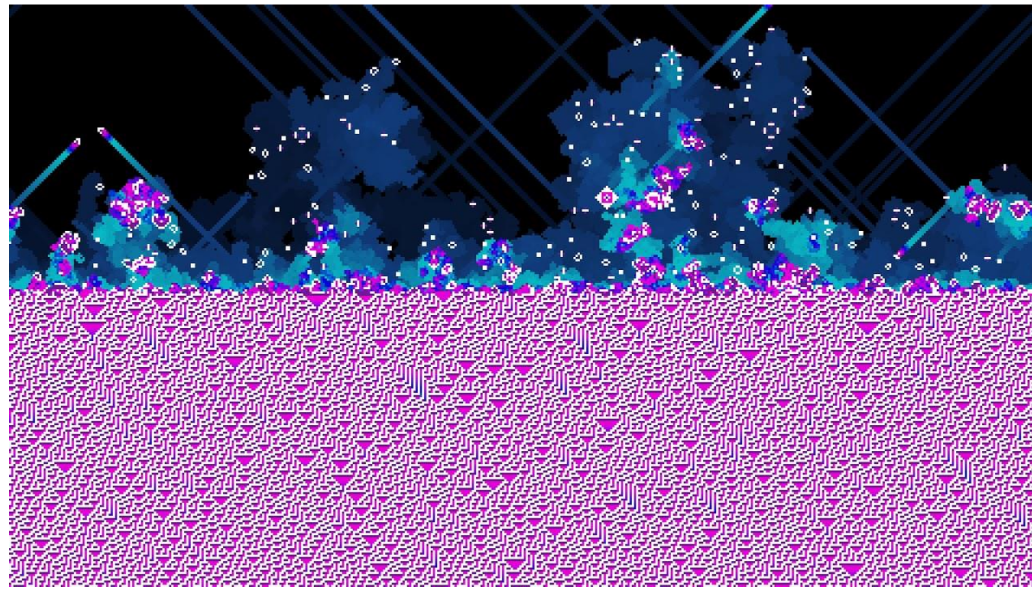
Chaos

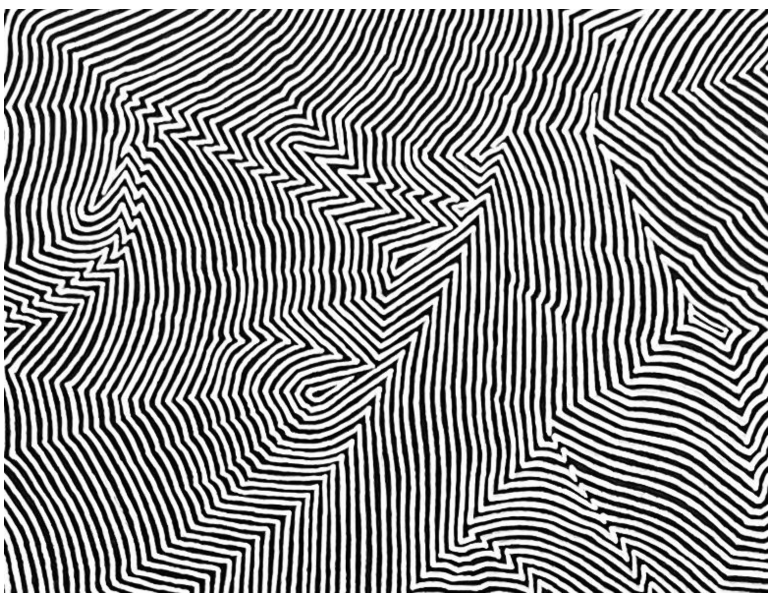
“When the present determines the future but the approximate present does not approximately determine the future.”
Edward Lorenz



https://en.wikipedia.org/wiki/Chaos_theory

Cellular Automata Rule 30





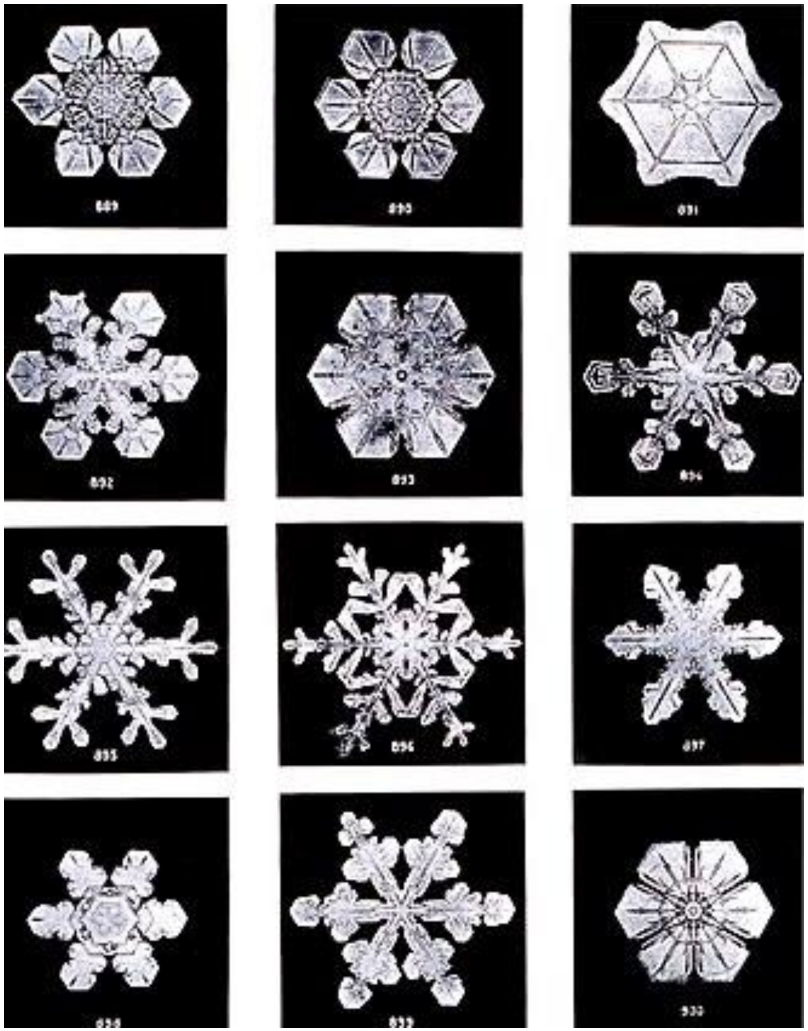
https://github.com/elliotwaite/rule-30-and-game-of-life

https://writings.stephenwolfram.com/2019/10/announcing-the-rule-30-prizes/

https://www.geograph.org.uk/photo/5398519

https://youtu.be/IK7nBOLYzdE

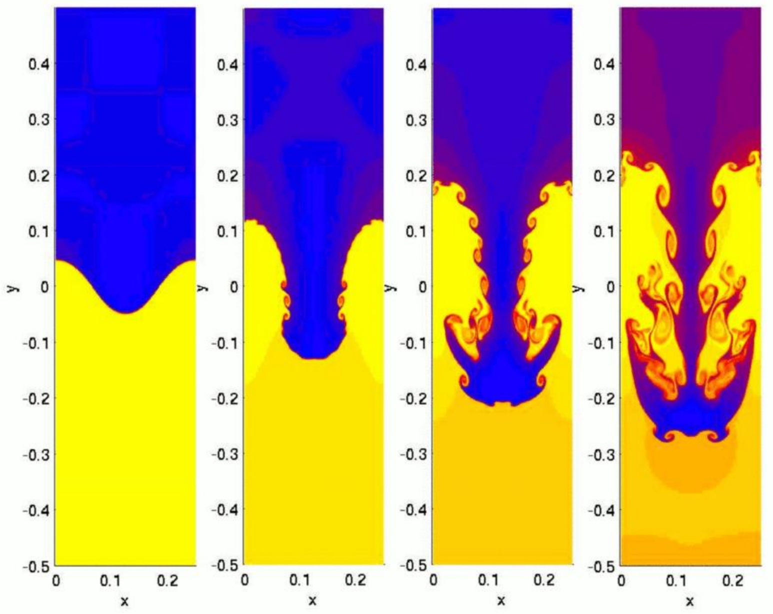
Emergence

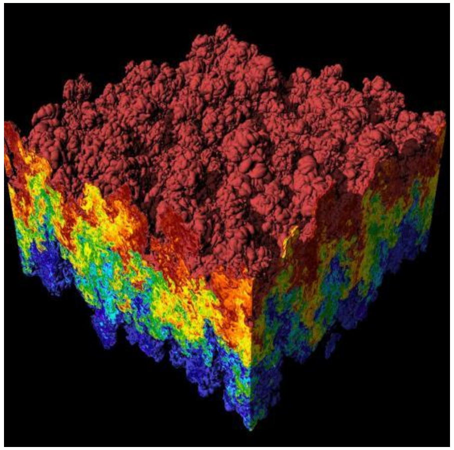


Termite mound

https://en.wikipedia.org/wiki/Emergence

Instability and Tipping Point





https://en.wikipedia.org/wiki/Rayleigh%E2%80%93Taylor_instability

Time-series variation and error

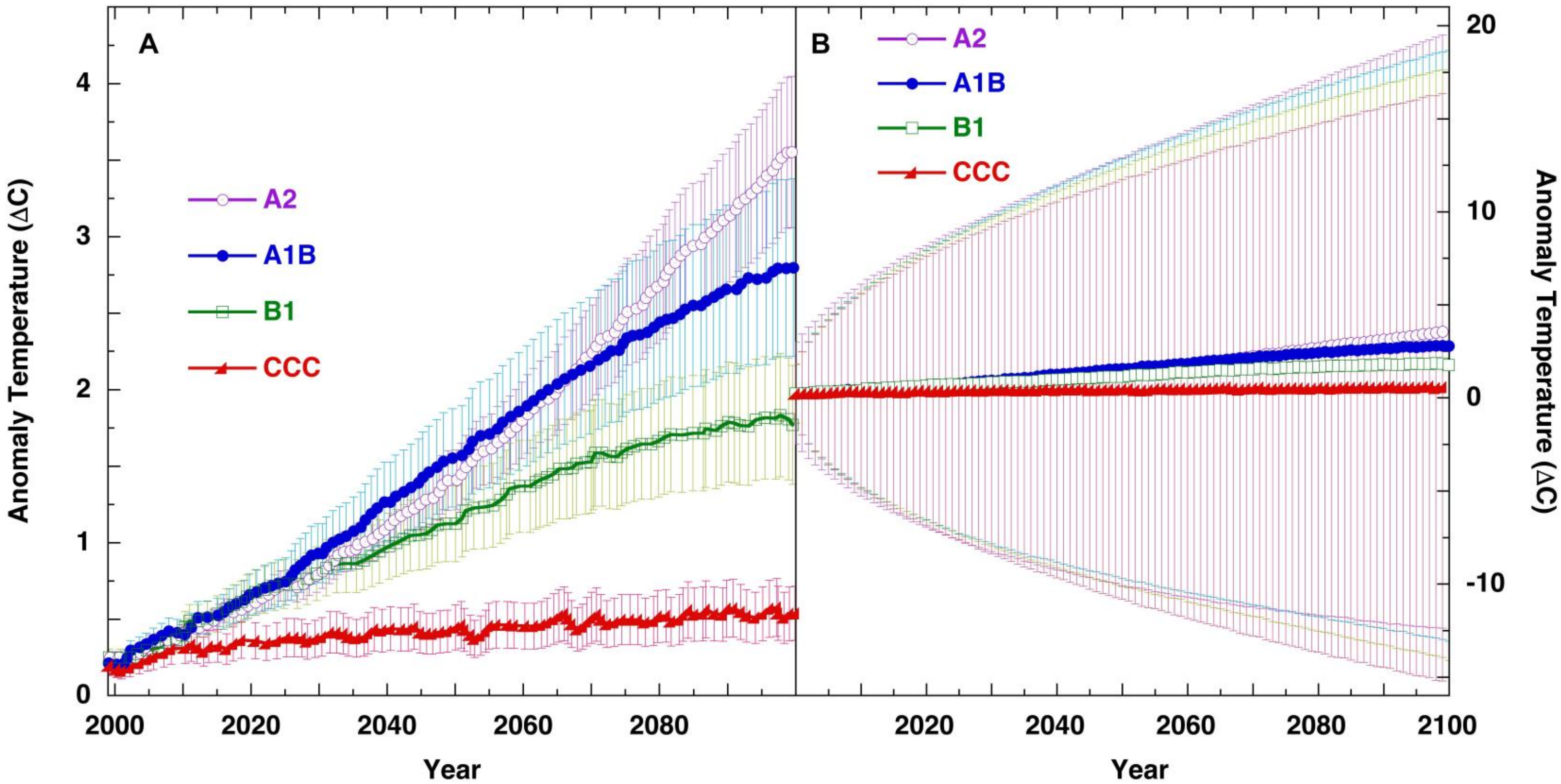
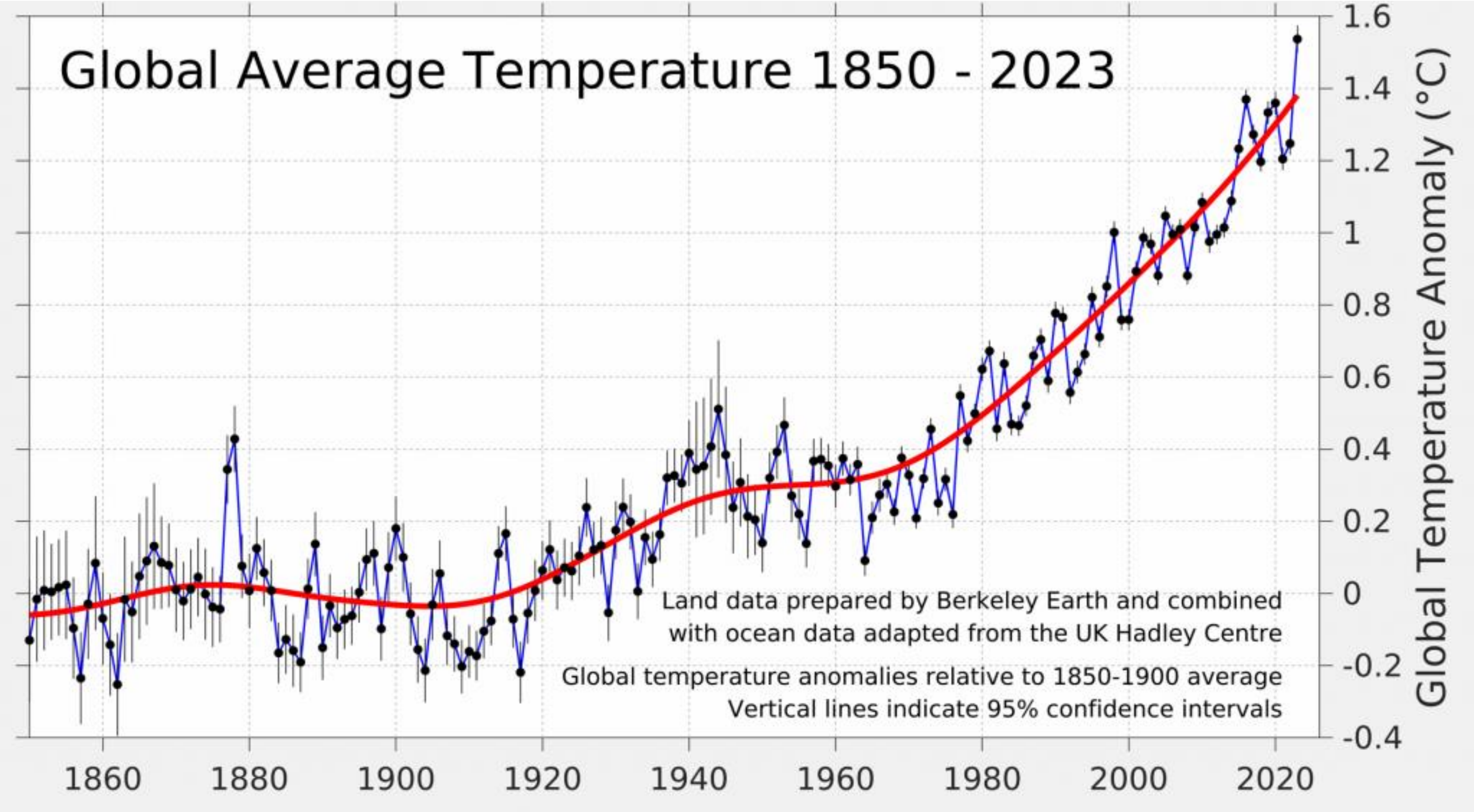


Figure 6. Panel **(A)**, SRES scenarios from IPCC 4AR WGI Figure SPM.5 ([IPCC, 2007](#)),
with uncertainty bars representing,
"the±1 standard deviation range of individual model annual averages."

https://www.frontiersin.org/journals/earth-science/articles/10.3389/feart.2019.00223/full

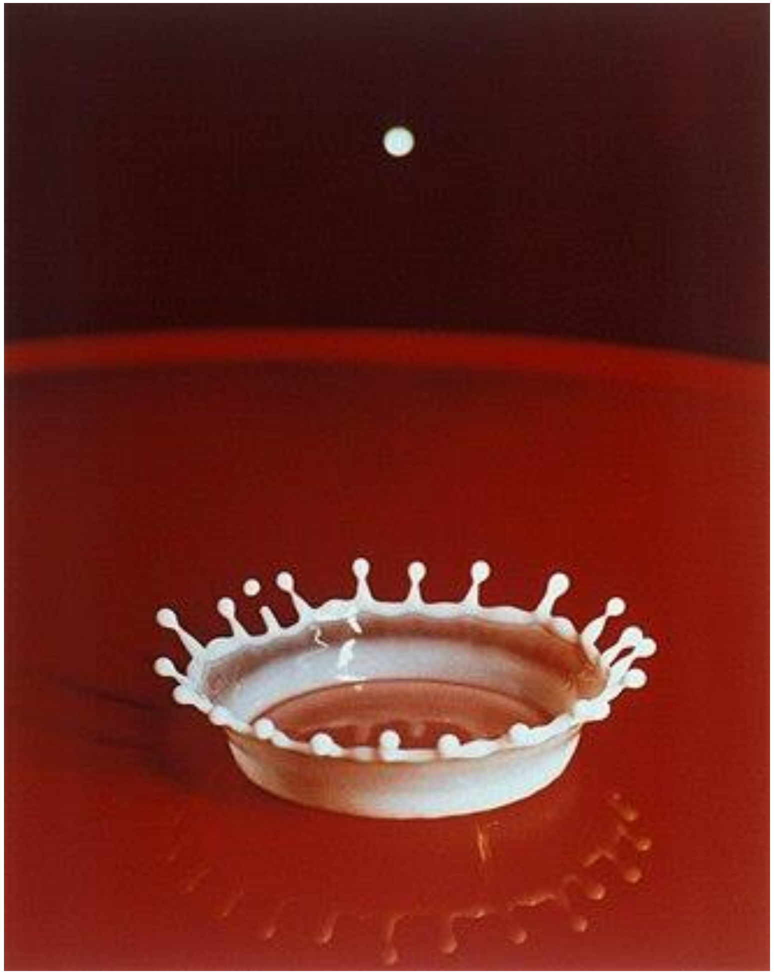
https://www.frontiersin.org/files/Articles/452488/feart-07-00223-HTML-r2/image_m/feart-07-00223-g006.jpg

Variations and Errors



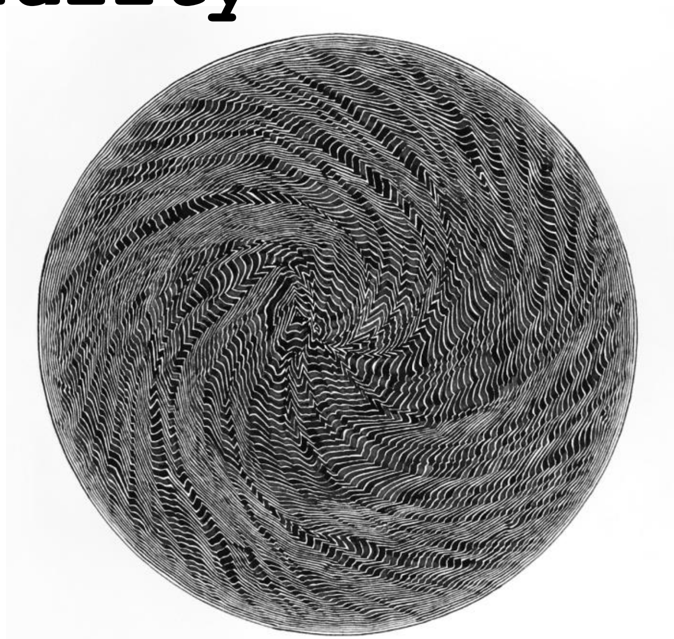
https://berkeleyearth.org/global-temperature-report-for-2023/

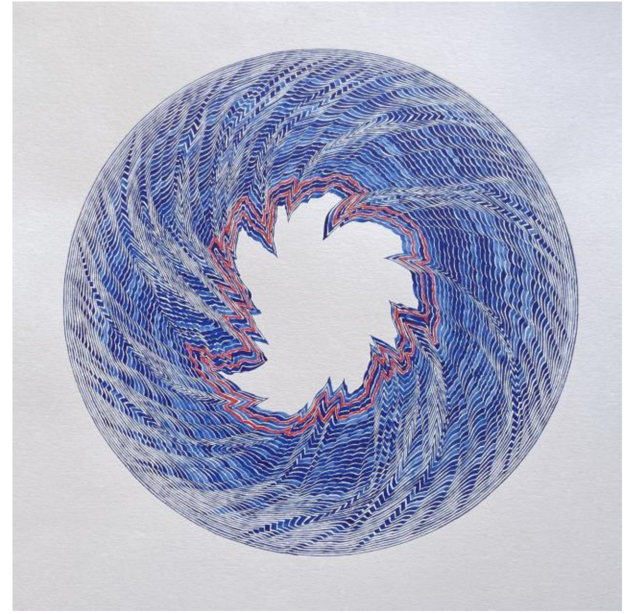
Symmetry Breaking

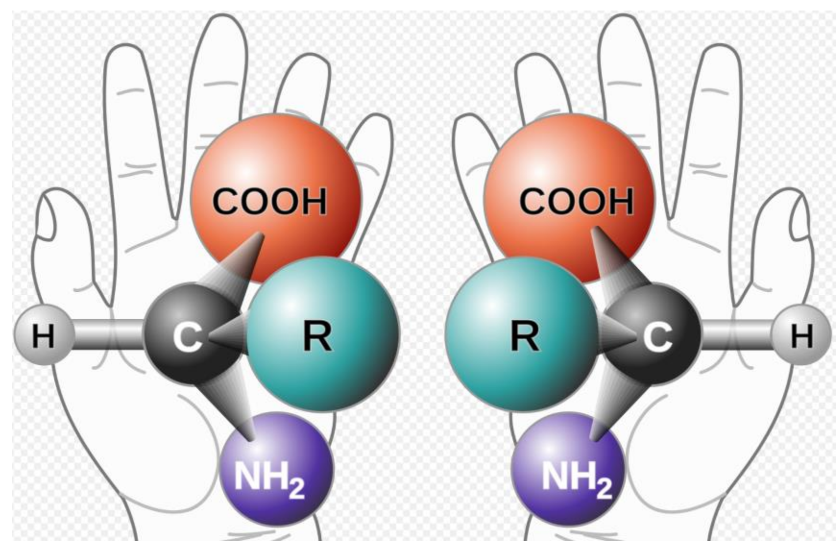


https://en.wikipedia.org/wiki/Milk_Drop_Coronet

Chirality



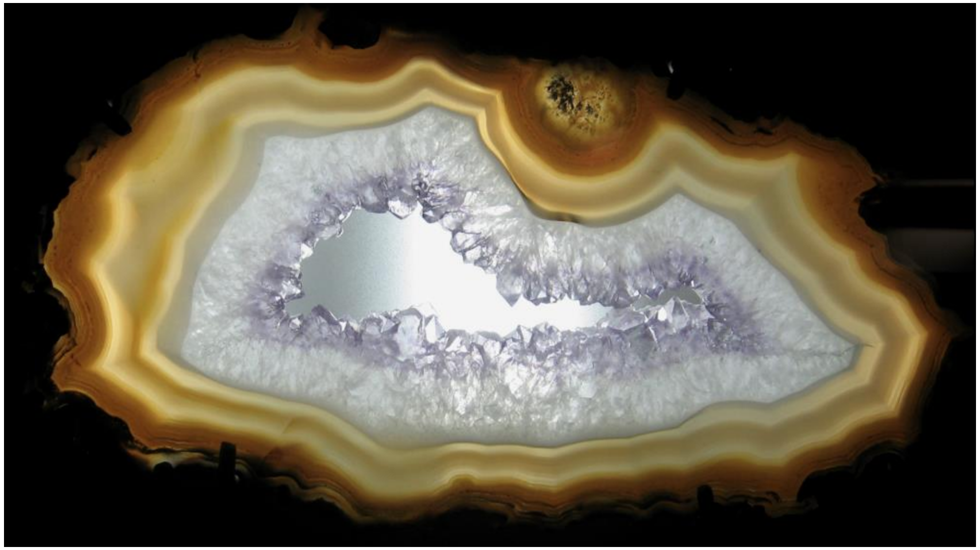


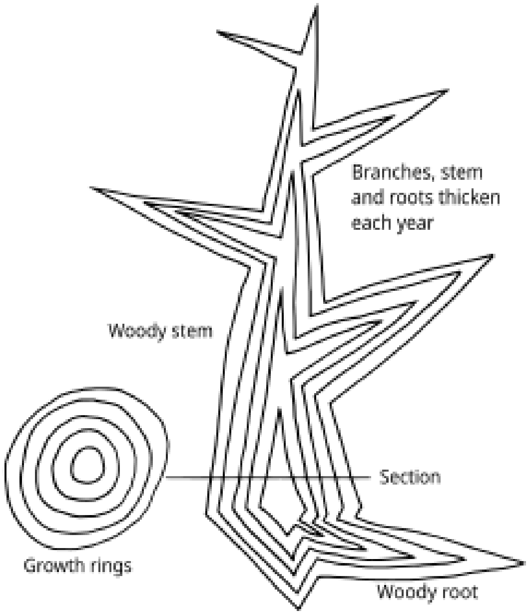


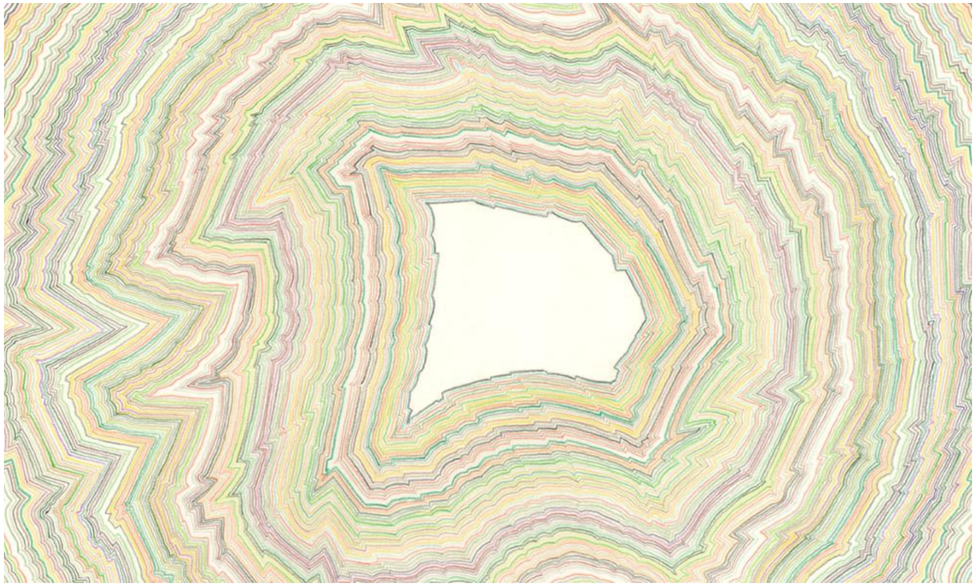
https://www.iucr.org/news/newsletter/volume-32/number-3/eternal-chirality

https://en.wikipedia.org/wiki/Chirality_(chemistry)

Layers







https://commons.wikimedia.org/wiki/Category:Geodes

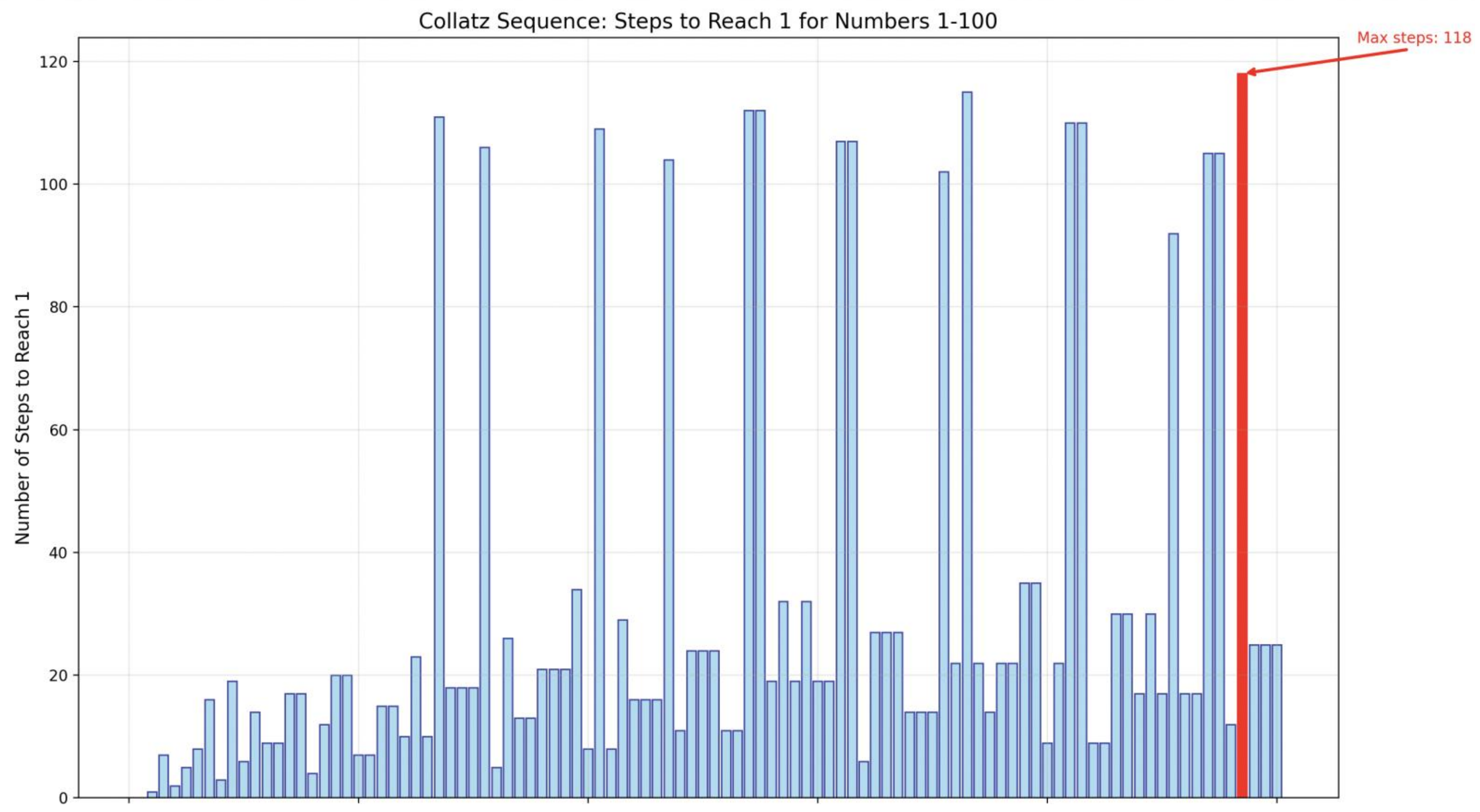
https://en.wikipedia.org/wiki/Dendrochronology

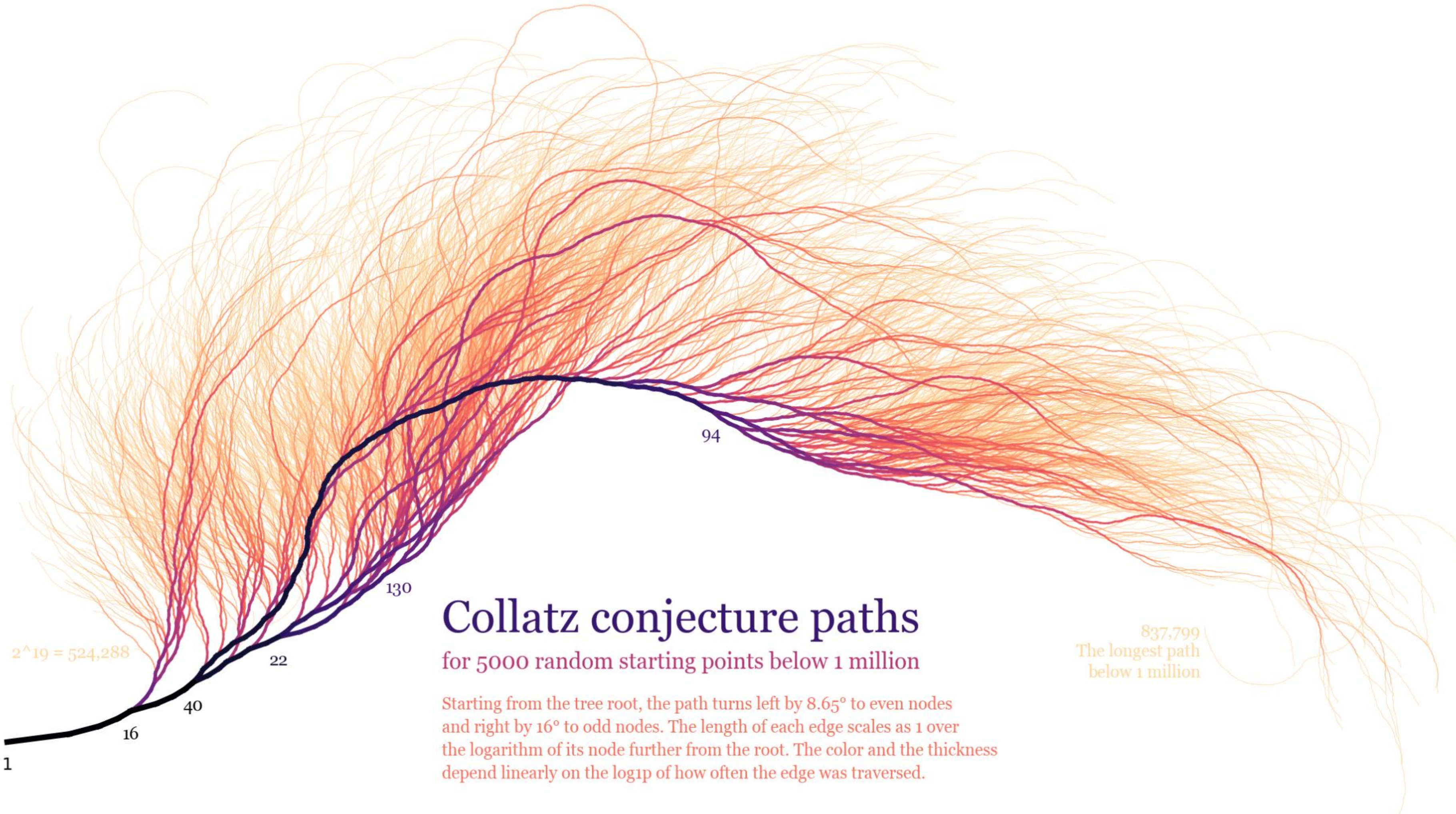
Molecular Dynamics

https://projects.volkamerlab.org/teachopencadd/talktorials/T020_md_analysis.html

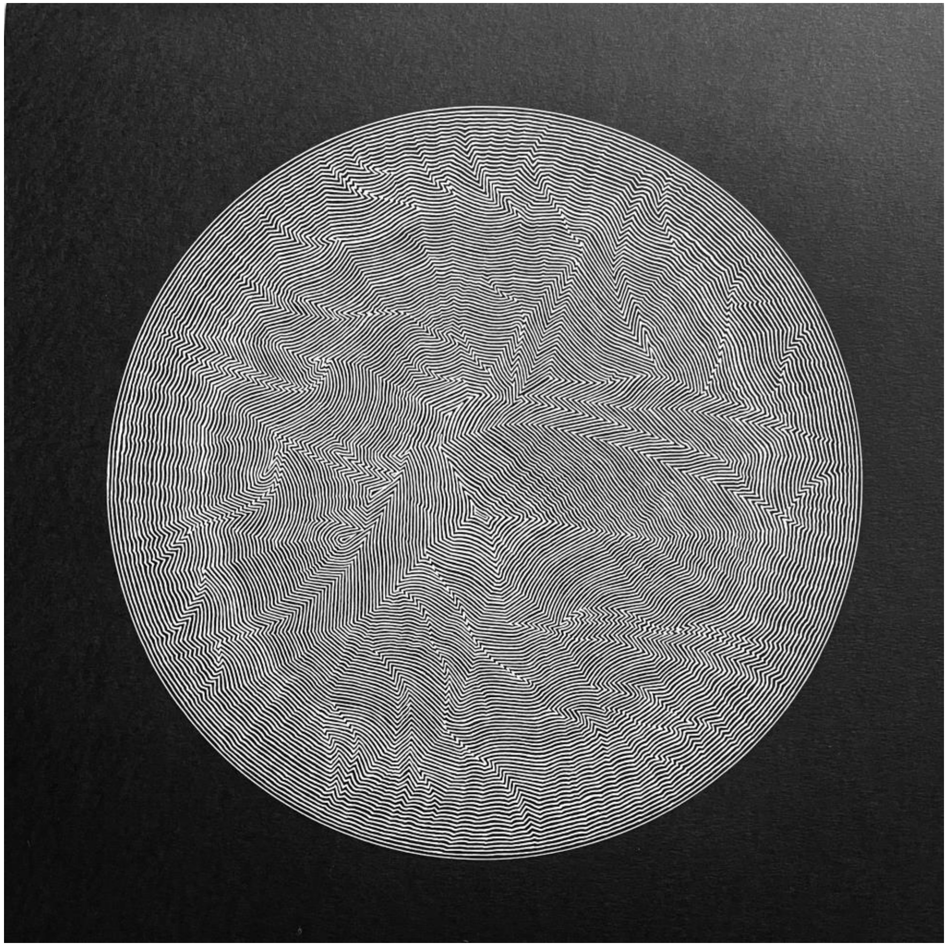
Algorithm

```
print("Rules:")
print("- If the number is even, divide by 2")
print("- If the number is odd, multiply by 3 and add 1")
print("- Keep going until you reach 1")
print("- Then pick a new number or stop")
print("- Range: 1 to 100")
print()
```




Collatz conjecture paths
for 5000 random starting points below 1 million
Starting from the tree root, the path turns left by 8.65° to even nodes and right by 16° to odd nodes. The length of each edge scales as 1 over the logarithm of its node further from the root. The color and the thickness depend linearly on the log1p of how often the edge was traversed.
2^19 = 524,288
837,799
The longest path below 1 million
1
16
40
22
130
94

Conversation with ChatGPT



Kirsten : May I show you a picture of an artwork and ask you to respond to it ?

ChatGPT said:

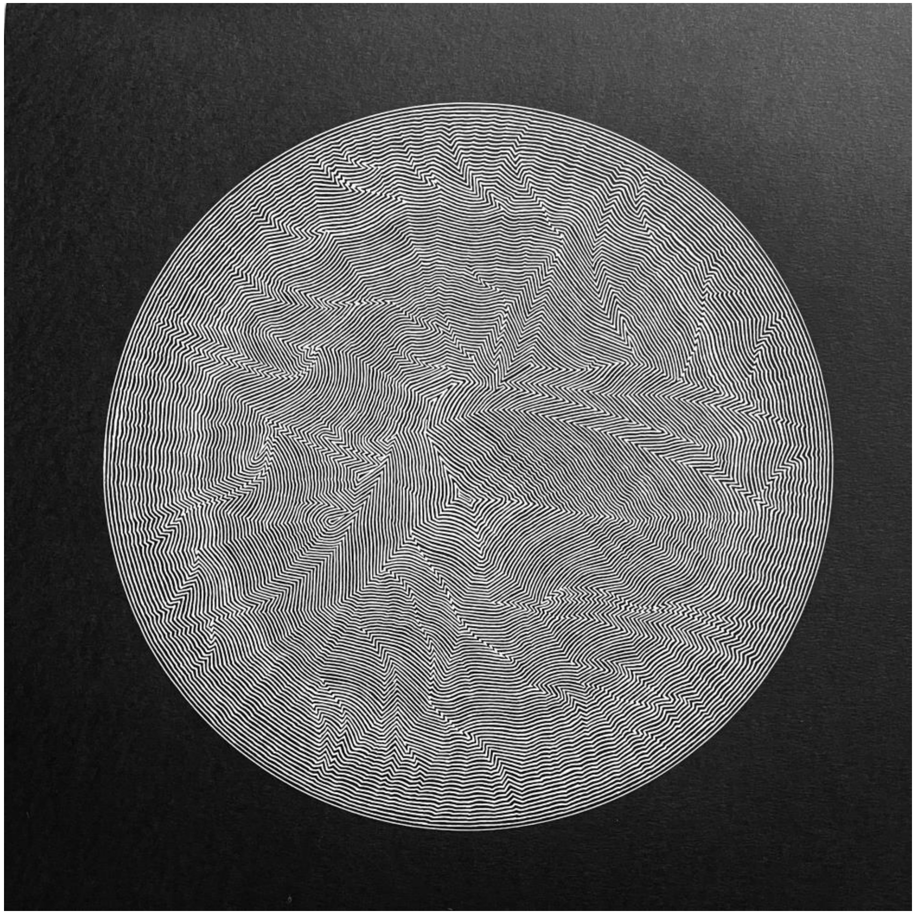
Absolutely — please go ahead and upload the image of the artwork. Once I've seen it, feel free to ask your question or let me know how you'd like me to respond (e.g., interpretation, critique, title suggestion, etc.).

Kirsten : interpretation please

ChatGPT said:

Thank you for sharing the image — it's a striking piece.

Conversation with ChatGPT

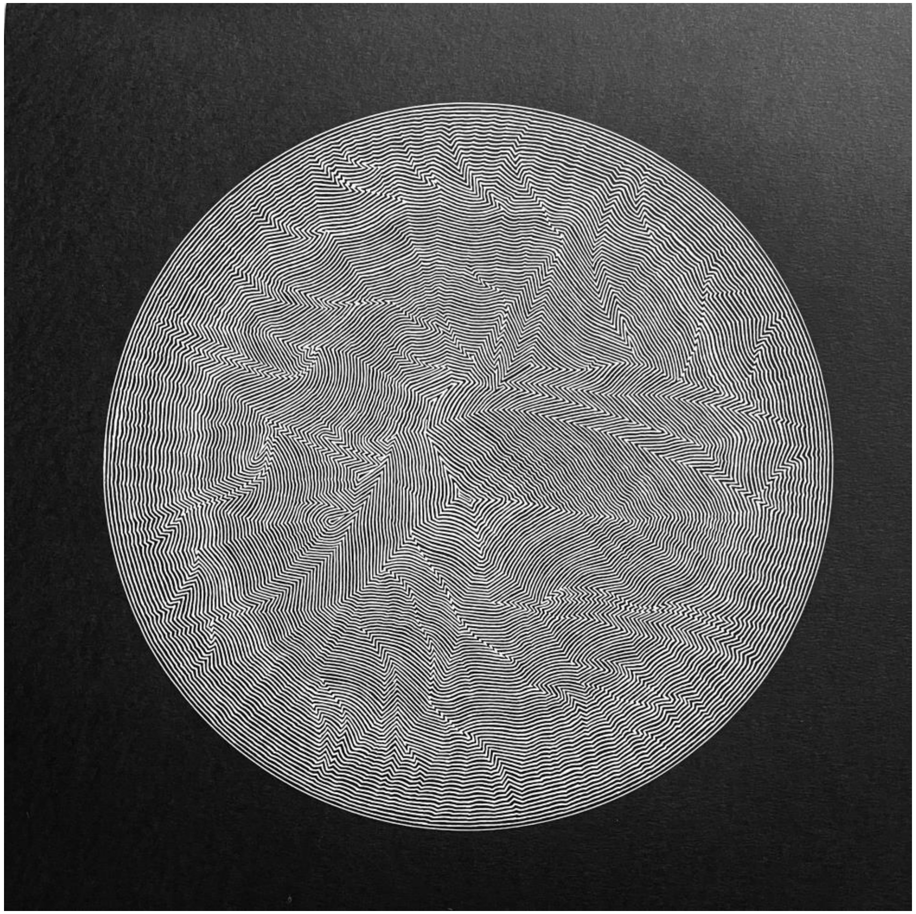


ChatGPT said: Here's a layered interpretation, moving from the visual and formal to the conceptual:

1. Visual/Formal Analysis

- **Line and Movement: The tightly packed, undulating white lines create a powerful sense of vibration and movement, as though the image is *pulsing* or *breathing*. The visual rhythm shifts as you move across the circle, with areas of turbulence and calm, suggesting a dynamic inner life.**
- **Structure and Disruption: At first glance, the lines appear evenly spaced, but closer inspection reveals purposeful distortions — zigzags, bends, and convergences. These disruptions interrupt any sense of symmetry and push the viewer to look closer, to seek the underlying pattern — if there is one.**
- **Circular Form: The circle often symbolizes unity, wholeness, or the cosmos. In this case, it could be read as a *bounded field of energy*, a closed system undergoing complex internal transformations.**

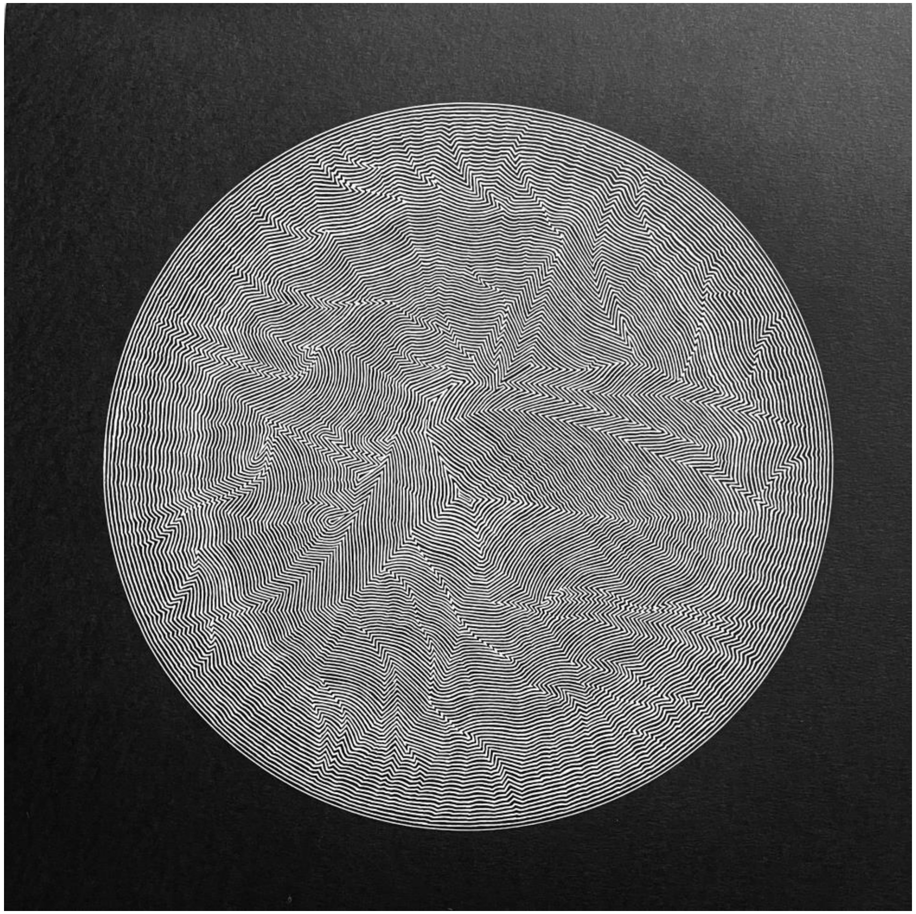
Conversation with ChatGPT



2. Emotional/Tonal Reading

- **There's an almost hypnotic quality to the piece — it invites prolonged contemplation but resists easy resolution.**
- **The interplay between order and chaos creates emotional tension: the comfort of repetition is constantly unsettled by unexpected deviations.**
- **This tension might evoke a sense of instability or flux, as though the image is a snapshot of something that's constantly in the process of becoming.**

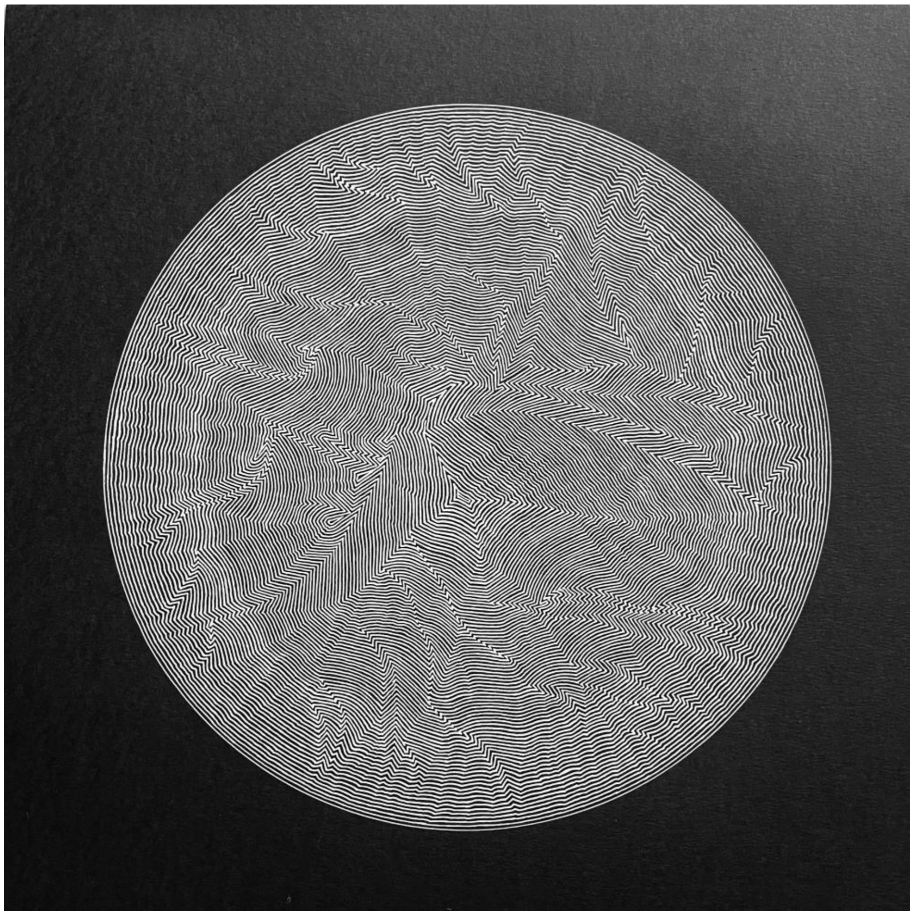
Conversation with ChatGPT



3. Conceptual/Thematic Possibilities

- Error and Precision: The work feels simultaneously algorithmic and human — as if it could have been plotted by a machine but *with a glitch*, or perhaps made by hand with extreme discipline and intentional variation. This tension could reflect themes of *imperfection within systems*, or *human presence within the mechanical.*
- Topography of the Mind or Sound: The forms could be interpreted as a kind of map — neural, emotional, or sonic. It recalls topographic maps, fingerprint whorls, or soundwaves distorted in transmission.
- Time and Process: It's also possible to read the image as a record of time, with each line like a sedimentary layer — a slow accumulation of decisions, gestures, or iterations.

Conversation with ChatGPT



Chat GPT said : If you'd like me to pair it with a text/poem, I'd be happy to help.

🌀 **1. Original Poem – Written in Response to the Artwork**

"Field of Deviations"
by ChatGPT, in response to your image

Within the circle, nothing stays still—
a hundred lines remember straightness,
then forget.

They wander like thought under pressure,
zigzagging in silence,
tracing the seams of some unseen logic.

A whisper of order—
then rupture,
a wave collapsed into another.

Is this how a mind looks
when it's almost calm?

Is this how time curls
before it breaks?

Conversation with ChatGPT

Kirsten asked: Please could you respond to this scientific diagram

Chat GPT said : This circular line diagram could be read scientifically as a kind of **contour map**, **interference pattern**, or **data visualization**, though without a legend or axis labels, it resists a direct “this-means-that” reading. Here’s a breakdown of possible interpretations through a scientific lens:

1. Visual Structure

- The symmetrical yet imperfect distribution could point to natural systems that have both **order** and **chaotic perturbations**.

2. Possible Scientific Analogues

- **Topography / Geophysics**: Could resemble a stylised elevation map of a mountainous or fractured terrain, with peaks and valleys encoded by line density and curvature.

3. Scientific-Philosophical Reading

It embodies the beauty of emergent patterns in complex systems: whether in weather systems, geological processes, or human-made algorithms, these structures are byproducts of rules interacting over time.

Developing AtPoE program with AI

AtPoE program

Curve Separation

8.00

0.00 15.00

Segment Length

3.00

1.00 15.00

Minimum Inter-Curve

1.00

1.00 4.00

Canvas Size

1000

🎯 Actions

+ Add Bundle | 🗑️ Clear All | ⏹️ Stop

📚 Bundle History

1. Custom Line (3 curves)
2. Custom Line (2 curves)
3. Custom Line (2 curves)

Parameters Used:

Curves: 2

Error: 5.4

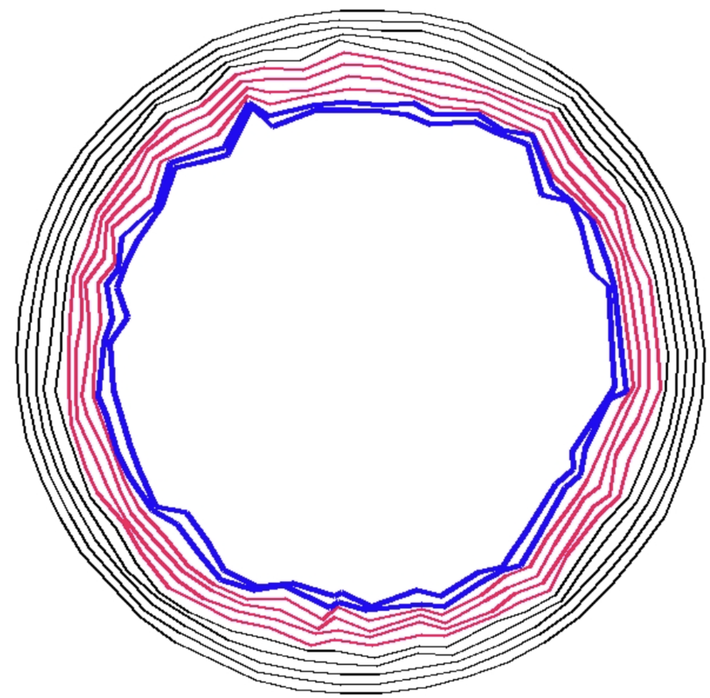
Separation: 8.0

Segment Length: 3.0

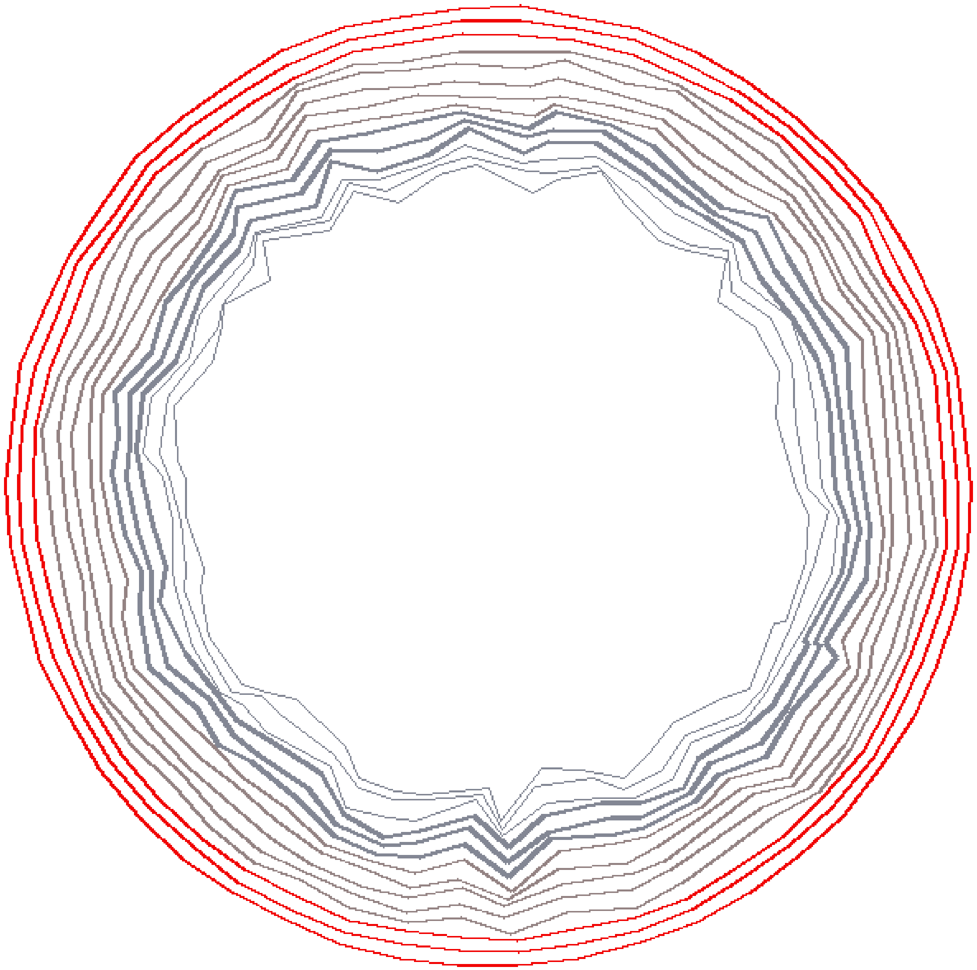
Min Inter-Curve: 1.0

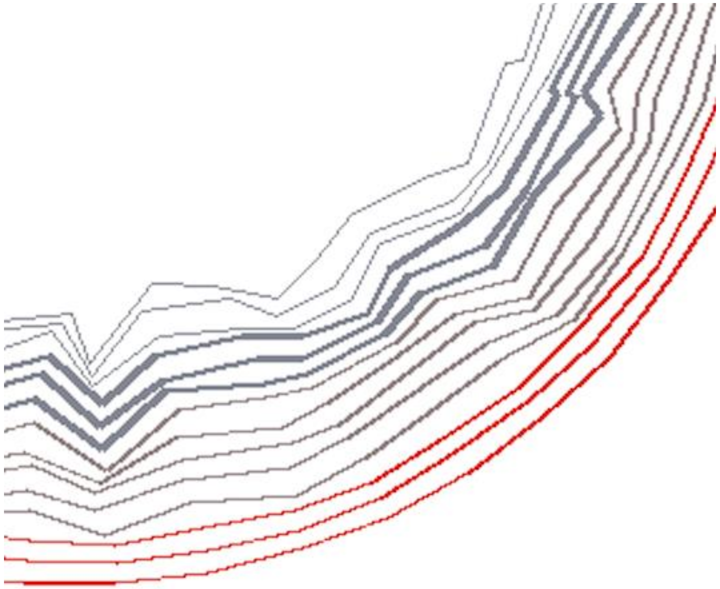
Width: 5.0

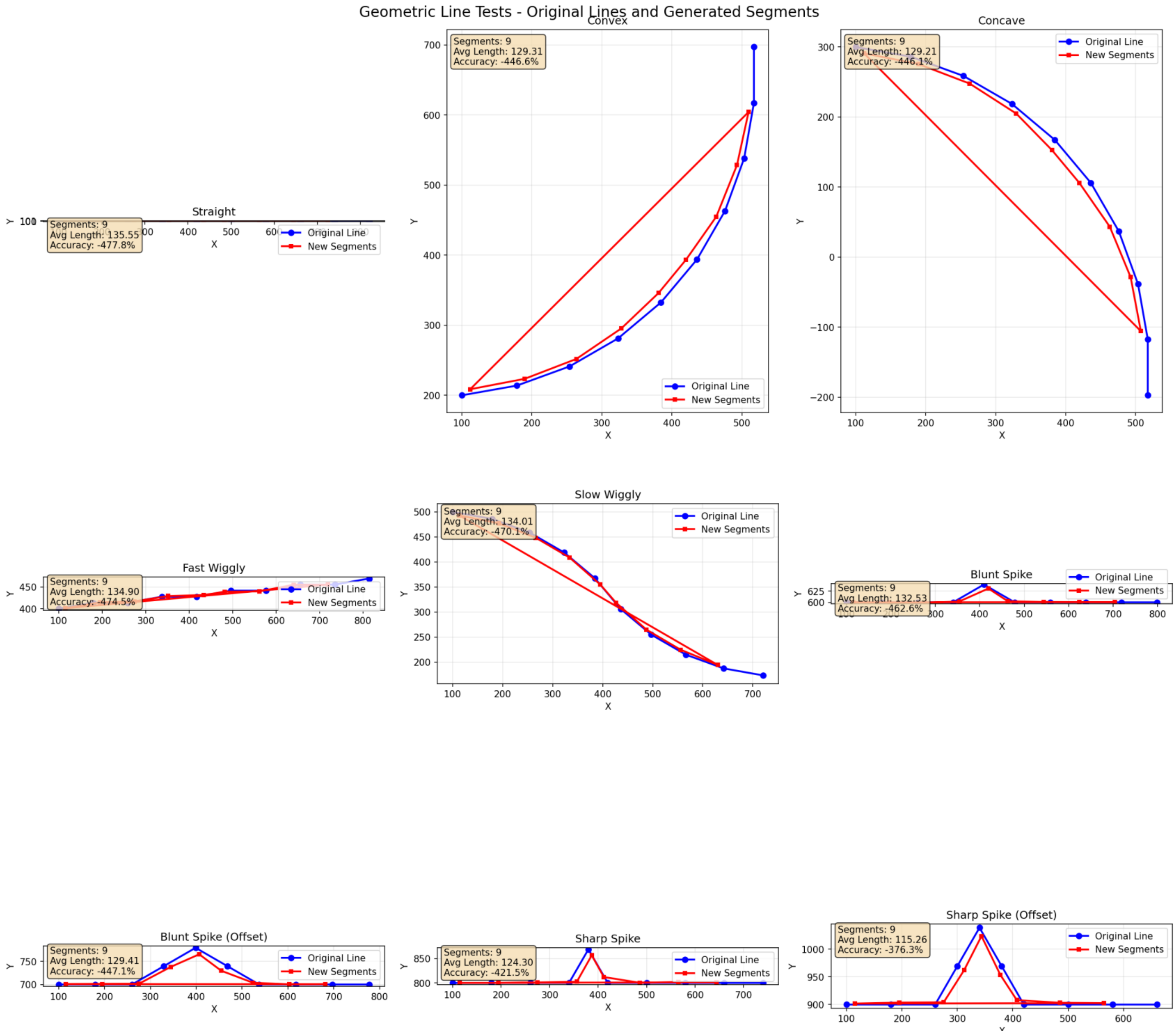
Color: #230af9



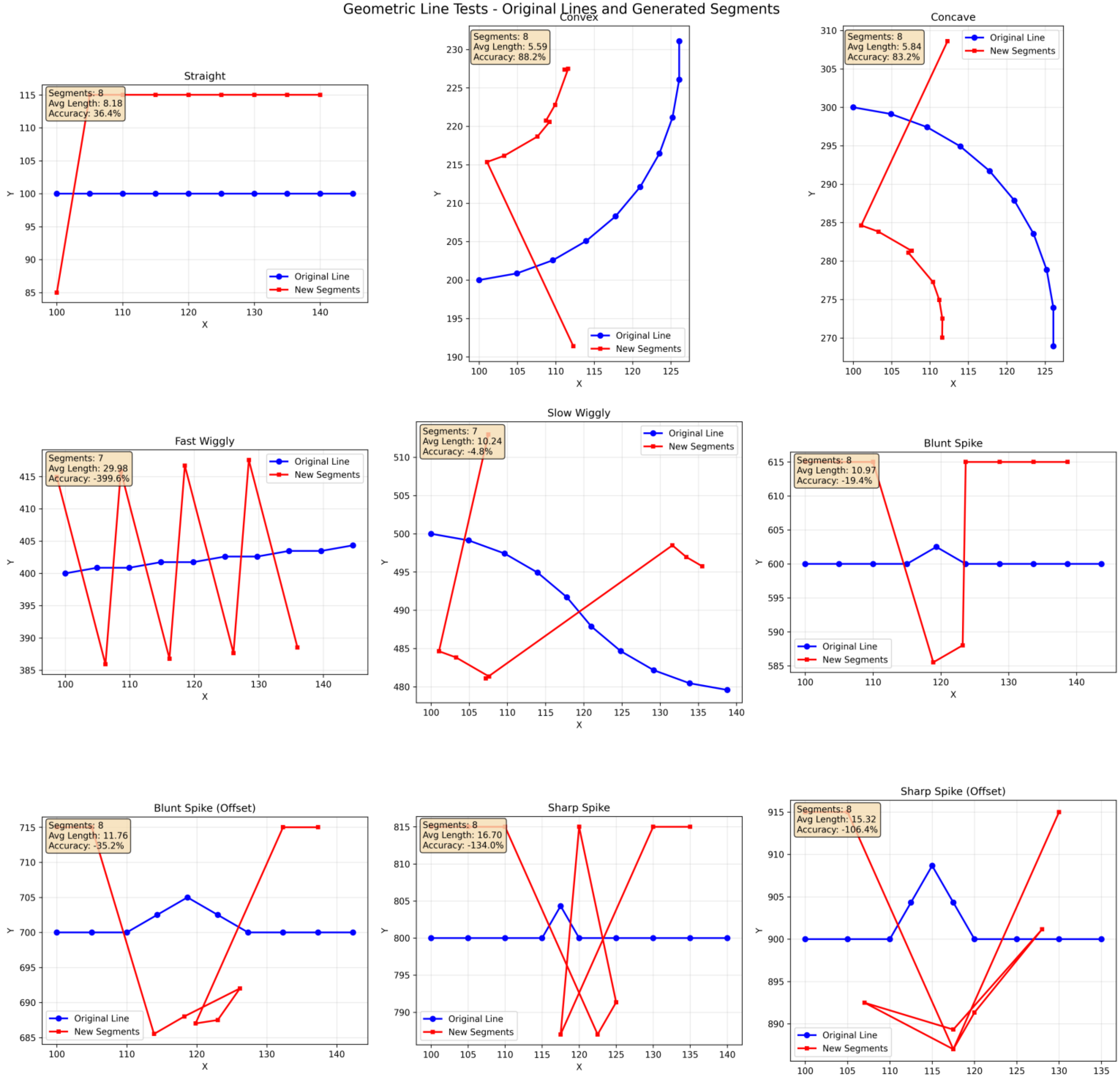
Atpoe_curves_4



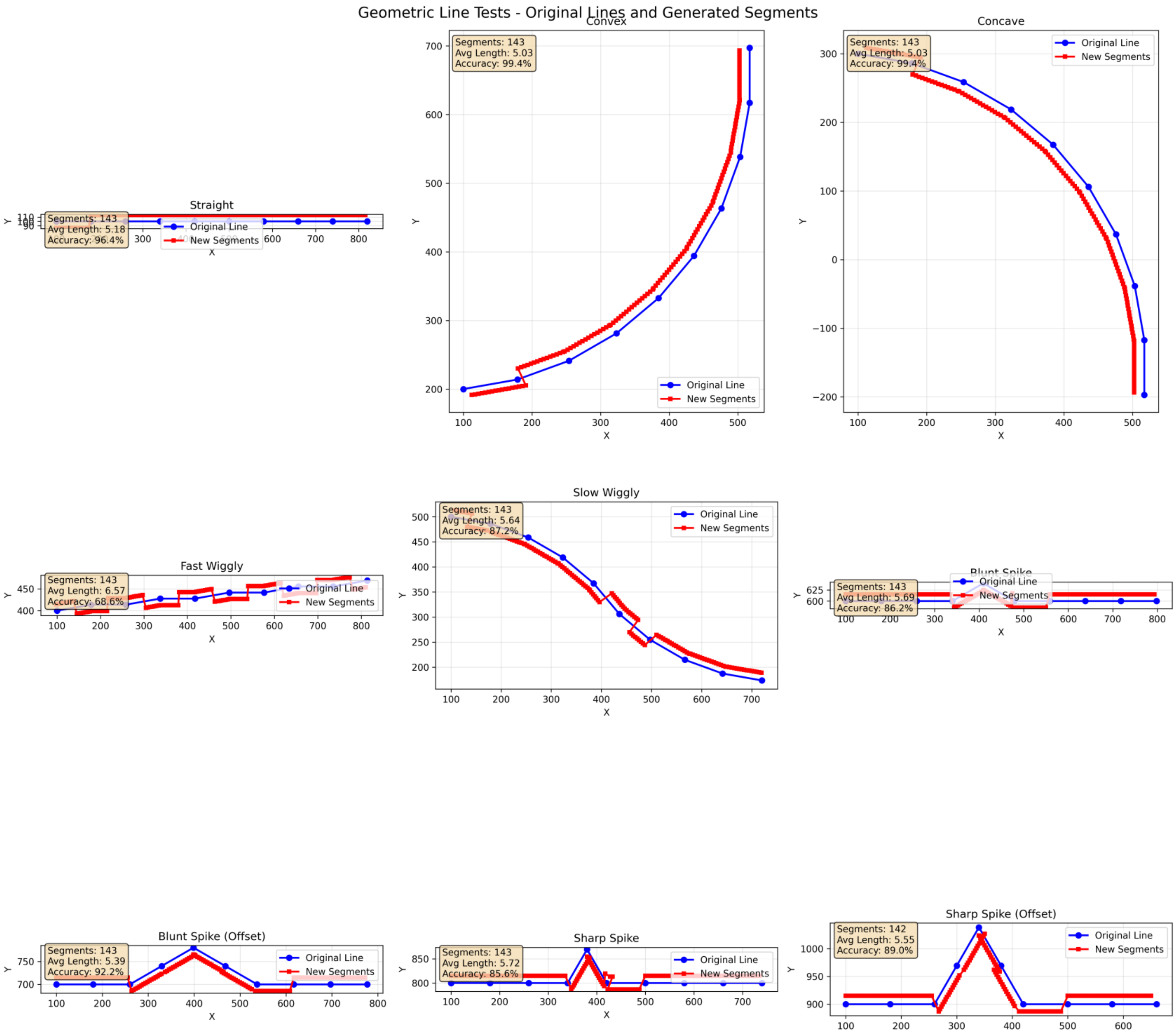


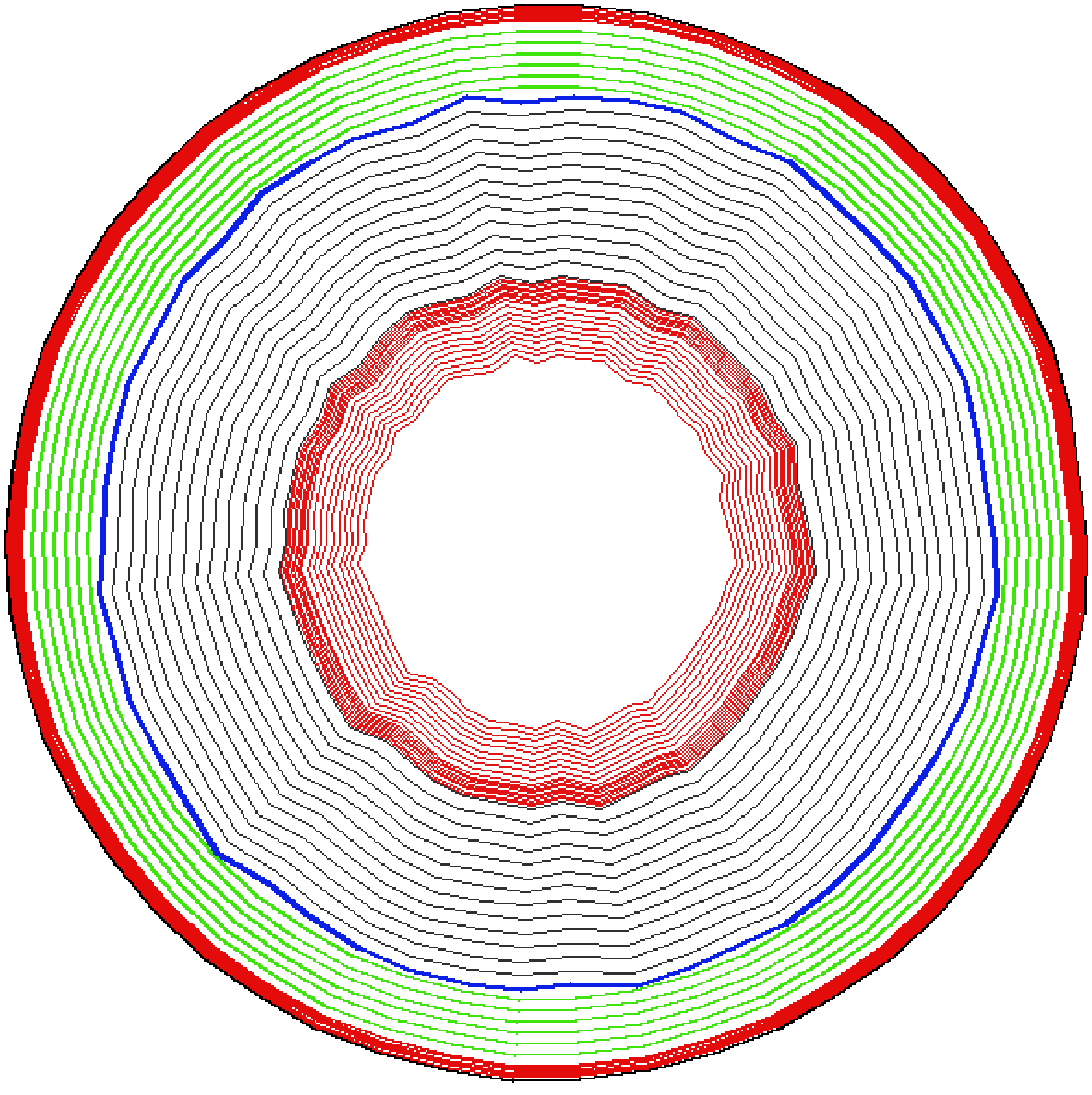


Geometric Line Tests - Original Lines and Generated Segments
Straight
Segments: 9
Avg Length: 135.55
Accuracy: -477.8%
Convex
Segments: 9
Avg Length: 129.31
Accuracy: -446.6%
Concave
Segments: 9
Avg Length: 129.21
Accuracy: -446.1%
Fast Wiggly
Segments: 9
Avg Length: 134.90
Accuracy: -474.5%
Slow Wiggly
Segments: 9
Avg Length: 134.01
Accuracy: -470.1%
Blunt Spike
Segments: 9
Avg Length: 132.53
Accuracy: -462.6%
Blunt Spike (Offset)
Segments: 9
Avg Length: 129.41
Accuracy: -447.1%
Sharp Spike
Segments: 9
Avg Length: 124.30
Accuracy: -421.5%
Sharp Spike (Offset)
Segments: 9
Avg Length: 115.26
Accuracy: -376.3%
Original Line
New Segments
X
Y



Geometric Line Tests - Original Lines and Generated Segments
Straight
Segments: 8
Avg Length: 8.18
Accuracy: 36.4%
Convex
Segments: 8
Avg Length: 5.59
Accuracy: 88.2%
Concave
Segments: 8
Avg Length: 5.84
Accuracy: 83.2%
Fast Wiggly
Segments: 7
Avg Length: 29.98
Accuracy: -399.6%
Slow Wiggly
Segments: 7
Avg Length: 10.24
Accuracy: -4.8%
Blunt Spike
Segments: 8
Avg Length: 10.97
Accuracy: -19.4%
Blunt Spike (Offset)
Segments: 8
Avg Length: 11.76
Accuracy: -35.2%
Sharp Spike
Segments: 8
Avg Length: 16.70
Accuracy: -134.0%
Sharp Spike (Offset)
Segments: 8
Avg Length: 15.32
Accuracy: -106.4%
Original Line
New Segments
X
Y


Geometric Line Tests - Original Lines and Generated Segments
Straight
Segments: 143
Avg Length: 5.18
Accuracy: 96.4%
Convex
Segments: 143
Avg Length: 5.03
Accuracy: 99.4%
Concave
Segments: 143
Avg Length: 5.03
Accuracy: 99.4%
Fast Wiggly
Segments: 143
Avg Length: 6.57
Accuracy: 68.6%
Slow Wiggly
Segments: 143
Avg Length: 5.64
Accuracy: 87.2%
Blunt Spike
Segments: 143
Avg Length: 5.69
Accuracy: 86.2%
Blunt Spike (Offset)
Segments: 143
Avg Length: 5.39
Accuracy: 92.2%
Sharp Spike
Segments: 143
Avg Length: 5.72
Accuracy: 85.6%
Sharp Spike (Offset)
Segments: 142
Avg Length: 5.55
Accuracy: 89.0%
Original Line
New Segments
X
Y



AtPoE

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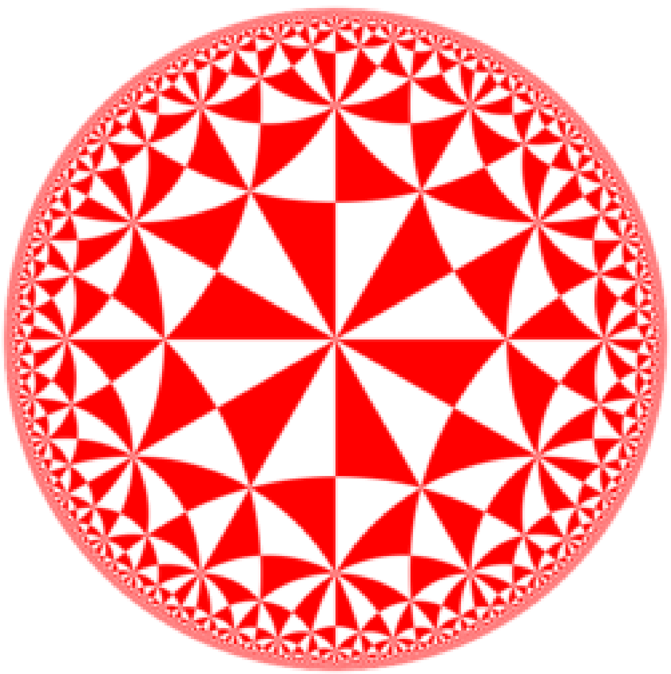
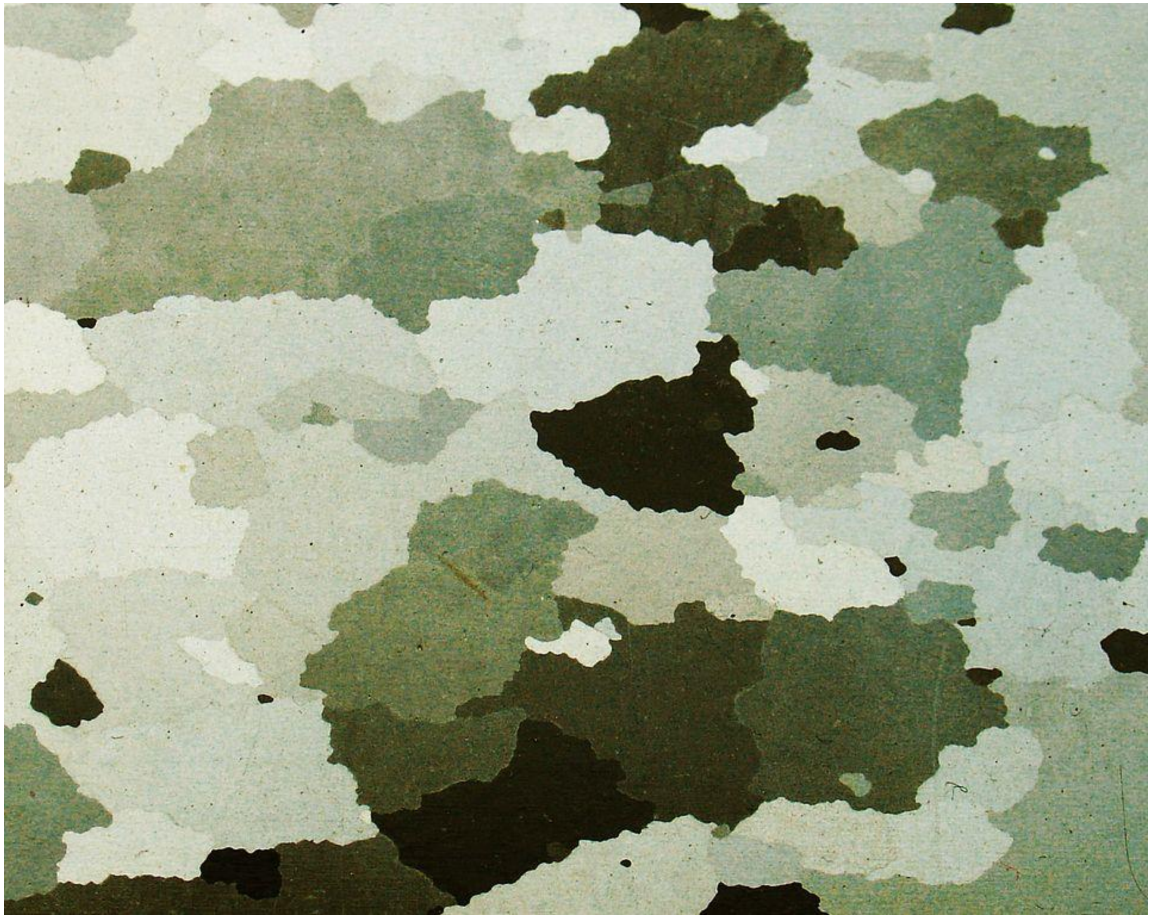
Maurits Escher

- https://en.wikipedia.org/wiki/M._C._Escher
- https://en.wikipedia.org/wiki/Circle_Limit_III



Steel Grains



https://commons.wikimedia.org/wiki/File:Grain-oriented_electrical_steel_(grains).jpg

- https://en.wikipedia.org/wiki/Personal_equation
- https://en.wikipedia.org/wiki/Emergence